

MkatabaWatch

Public Contracts, Public Evidence (Mikataba ya Umma, Ushahidi wa Umma)

A working civic accountability platform connecting verified official Tanzanian public procurement data with citizen on-the-ground evidence, powered by an objective, non-accusatory AI reconciliation engine.

Track: Transparency & Accountability	Live Data Source: Tanzania PPRA NeST Portal	Author / Developer: Ian Karanja
Repository: github.com/iankaranja13/mkatabawatch	Coverage: OCDS v1.1 National Feed (TZS 61.8B+ Tested)	Status: Working Proof of Concept

The Problem: The Physical Blindspot in Public Procurement

Government portals track contract paperwork and financial disbursements—not on-the-ground reality.

1. The Data Transparency Paradox

Tanzania's Public Procurement Regulatory Authority (PPRA) has made remarkable strides publishing thousands of contracts on the NeST portal. However, procurement systems naturally stop at the computer screen: they confirm tender awards and bank transfers, but cannot confirm if brick and mortar were laid.

2. The Ghost & Stalled Project Dilemma

Billions of shillings are allocated to critical public works—flood mitigation culverts, primary school dormitories, rural water treatment plants. When a contractor mobilizes late, abandons the trench, or uses substandard materials, months or years pass before official auditors arrive.

3. Citizens Have Eyes, but No Voice or Data

Local residents walk past unroofed classrooms and abandoned drainage culverts every day. Yet citizens rarely know the official contract terms: Who won the tender? What was the budget? When was the legal deadline? Without contract facts, citizen feedback is dismissed as unverified hearsay.

4. The 3-Year Retrospective Audit Lag

Traditional audits by the Controller and Auditor General (CAG) are meticulous but retrospective—often published 2 to 3 years after project funds have already been spent and contractors have demobilized.

The Core Failure: A severe disconnect between *de jure* contractual commitments reported in capital cities and *de facto* infrastructure delivered in rural and municipal communities.

Target Users: Closing the Loop Across 4 Key Ecosystems

MkatabaWatch is designed for collaboration between civic monitors, auditors, and civil society.

1. Citizens & Community Monitors Who they are: Local residents, youth champions, ward development committee members. Pain point: Lack access to official contract figures; cannot tell if stalled construction violates legal agreements. Value gained: Transparent access to contract scopes and an effortless mobile portal to submit geotagged photos in English or Swahili.	2. Civil Society Organizations (CSOs) Who they are: Accountability watchdogs (WAJIBU, HakiRasilimali, Twaweza, Policy Forum). Pain point: Field investigations require months of painstaking manual paper matching. Value gained: Automated AI discrepancy flagging that instantly prioritizes high-risk projects requiring on-site audit missions.
3. Government Oversight Bodies Who they are: PPRA, CAG, Parliamentary Public Accounts Committees, Regional Commissioners. Pain point: Inability to physically inspect tens of thousands of decentralized projects. Value gained: Crowdsourced, real-time ground telemetry that serves as an early-warning radar before projects fail.	4. Ethical Contractors & Procuring Entities Who they are: Registered civil works suppliers and local government authorities (LGAs). Pain point: Lumped into widespread public skepticism even when executing projects well. Value gained: Public verification of on-time milestone delivery, protecting vendor reputation and encouraging timely payment.

The Solution: The 4-Stage MkatabaWatch Reconciliation Loop

Pairing official open data with citizen telemetry and responsible AI discrepancy detection.

STAGE 1: Official OCDS Ingestion Ingests genuine OCDS v1.1 records from Tanzania PPRA NeST portal. Extracts OCID, buyer, vendor, contract value, award date, and contractual completion deadlines with 100% data authenticity.	STAGE 2: Community Evidence Capture Citizen monitors capture geotagged photos, observation categories (e.g. <i>Stalled</i>, <i>Incomplete</i>, <i>Defective</i>), GPS coordinates, and descriptions on 3G-friendly mobile interfaces.
STAGE 3: AI Reconciliation Engine Anthropic Claude 3.5 Sonnet / Heuristic engine compares contractual timelines with submitted field evidence under strict anti-accusation safety guardrails, producing structured findings.	STAGE 4: Human Verification Queue Flagged discrepancies enter an auditor review queue. Human investigators mark as <i>Verified</i> for audit, <i>Resolved</i>, or <i>Dismissed</i>, maintaining humans firmly in the loop.

Responsible AI Architecture: The AI engine never generates official procurement numbers and never accuses anyone of corruption or crimes. It strictly surfaces objective divergence between stated schedules and visible site progress.

Live Reality: 10 Verified Tanzanian Contracts (TZS 61.8 Billion)

Tested on real public procurement contracts spanning water, roads, health, and schools.

Project & Procuring Entity	Contractor & Value	Official Stated Terms	Observed Evidence & AI Status
Dar es Salaam Culverts & Bridges Dar es Salaam City Council	Crossworld Construction TZS 1,522,225,450	Construction of 4 box culverts & drift at Msumi and Kisukuru.	DISCREPANCY FLAGGED Excavator abandoned; flooded trenches; pipes dumped beside road.
Korogwe Water Treatment Plant Handeni Water Supply Authority	Tumaini Civil Works TZS 888,169,360	365 days water filtration tanks construction (ended July 2025).	DISCREPANCY FLAGGED Site deserted since August; unpoured vertical rebar corroding in rain.
Mtimbira Primary School Dormitory Malinyi District Council	Zidadu General Supplies TZS 82,195,600	30-day building materials delivery for 80 pupils (ended June 2024).	DISCREPANCY FLAGGED Timber and roofing missing 3 months late; walls unroofed; pupils displaced.
TARURA National HQ Supervision TARURA Dodoma	National Housing Corp TZS 560,451,500	Supervision of multi-story HQ in Njedengwa Investment Area.	CONSISTENT Perimeter hoarding up, tower crane active, basement concrete poured on schedule.
Ukerewe Referral Hospital Supervision Mwanza Regional Secretariat	Malk Consultants TZS 960,745,000	1,095 days hospital architectural design and construction supervision.	CONSISTENT Geotechnical soil drilling rigs and public EIA stakeholder meetings active.

Technical Excellence: Built for Real-World African Infrastructure

Engineered for low-bandwidth 3G devices, offline durability, and zero-setup deployment.

1. Low-Bandwidth 3G Optimization Total frontend asset footprint is under 60 KB with zero heavy JavaScript frameworks. Loads instantaneously even on rural 2G/3G connections.	3. Native English & Swahili i18n Seamless bilingual toggle across all 5 screens, filter dropdowns, and buttons, ensuring accessibility for local Tanzanian community monitors.	5. Audit-Grade Data Schema Normalized relational SQLite/Postgres schema maintaining strict provenance separation between official records and community telemetry.
2. Instant OCDS Inspector Modal Built-in sub-millisecond OCDS JSON inspector eliminates reliance on slow external government web servers while maintaining 100% data fidelity.	4. Dual-Mode AI Inference Supports live Anthropic Claude 3.5 Sonnet / OpenAI models via API, with an automatic deterministic heuristic engine for zero-credential offline evaluation.	6. Strict Multi-Field Validation Mandatory photo uploads, GPS coordinates, observation types, and descriptions prevent frivolous spam and ensure evidentiary integrity.

60 KB	11	100%	0
Asset Payload (3G Fast)	API Endpoints Registered	OCDS v1.1 Standard Compliant	False Accusation Policy

Potential Impact: Transforming Public Infrastructure Delivery

Moving from retrospective post-mortems to proactive, early-warning civic accountability.

1. Drastic Reduction in Audit Cycles Compresses the discovery cycle of abandoned or delayed public projects from 2-3 years down to days or weeks, enabling interventions before contractors leave the site.	2. Safeguarding Public Treasury Helps procurement entities halt milestone disbursements when verified evidence proves materials have not been delivered, protecting billions in public funds.
3. Rebuilding Citizen Trust Transforms public cynicism into constructive monitoring by providing citizens with real contract data and an institutional channel that gets formally reviewed.	4. Pan-African Scalability Because OCDS is an international data standard adopted across Africa, MkatabaWatch can seamlessly expand to Kenya, Uganda, Ghana, Nigeria, and Rwanda.

"Transparency is not just about publishing PDF contracts on government portals; transparency is ensuring that the dispensary promised on paper actually dispenses medicine to the community."

Strategic Roadmap & Call to Action

From hackathon proof-of-concept to institutional civic infrastructure.

Phase 1: Working Core (Current) • Live NeST OCDS REST API Ingestion • 5 Interactive Core Screens • AI Reconciliation (Claude 3.5 + Rules) • Human Verification Triage Queue • Native English & Swahili Localization	Phase 2: Last-Mile Reach (Q4 2026) • USSD / SMS gateway integration for offline rural feature phones • WhatsApp reporting chatbot • Automated photo EXIF forensics (timestamp & location verification)	Phase 3: Institutional Scale (2027) • Direct integration with CAG & PPRA audit dispatch systems • Satellite imagery analysis for road & dam excavation verification • Expansion to Kenya, Uganda, & Ghana
--	---	--

Support MkatabaWatch in Championing Public Contract Accountability

Explore the code, test the live demo, and review our data notes on GitHub.

Repository: github.com/iankaranja13/mkatabawatch
Live API: <http://localhost:8000/docs>